PaperSonar

Project proposal

Phase 1 design deliverable v2 | 10 November 2026

Purpose

PaperSonar proposes a Research & Innovation Intelligence Warehouse that links publications, preprints, datasets, research software and related outputs. It helps researchers compare directions using publication/output activity, citation cohorts, topic intersections, creator/institution participation, access and explicit output relationships. It does not prove scientific novelty.

Requirements / design

Deliver a reproducible, zero-cost local analytical foundation first; validate saved comparison dossiers with researchers only after the warehouse metrics are reliable.

Scope and caveats

OpenAlex, Crossref, arXiv, DataCite and Zenodo; PostgreSQL required baseline; Snowflake optional only with no-cost trial/educational allowance.

Source and architecture references

See docs/sources.md, docs/architecture/README.md, docs/requirements/incremental-ingestion-design.md, docs/requirements/course-requirements-matrix.md, and the target architecture diagram. Phase 2 and Phase 3 implementation has not started.